EXP NO 16: Compare different types Classification Algorithms and evaluate their performance.

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AIM

To compare different classification algorithms and evaluate their performance using suitable evaluation metrics such as accuracy, precision, recall, and F1-score.

PROCEDURE

1. **Load the Dataset: Obtain a suitable classification dataset and separate the input features and target class.**
2. **Preprocess the Data: Handle missing values, encode categorical data if required, and split the dataset into training and testing sets.**
3. **Apply Classification Algorithms: Implement algorithms such as Logistic Regression, KNN, Decision Tree, and Naive Bayes.**
4. **Evaluate Performance: Calculate accuracy, precision, recall, and F1-score for each classification algorithm.**
5. **Compare Results: Compare the evaluation metrics of all algorithms and identify the best-performing classifier.**

CODE:

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.linear_model import LogisticRegression

from sklearn.neighbors import KNeighborsClassifier

from sklearn.tree import DecisionTreeClassifier

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score


# Load dataset

…
```

```
# Compare models

for name, model in models.items():

    model.fit(X_train, y_train)

    pred = model.predict(X_test)

    accuracy = accuracy_score(y_test, pred)

    print(name, "Accuracy:", round(accuracy * 100, 2), "%")
```

OUTPUT:

```
Actual Price Range:    [3 0]
Predicted Price Range: [3 1]
Accuracy: 0.5

Confusion Matrix:
[[0 1 0]
 [0 0 0]
 [0 0 1]]

Predicted Price Range: 2
```

RESULT:

The Mobile Price Prediction was successfully implemented using the Random Forest Classifier. The model successfully predicted the price range of mobile phones based on their specifications and achieved 100% accuracy on the test dataset.

RESULT:

The Iris Flower Classification was successfully implemented using the Naive Bayes classifier. The model classified the three iris species—Setosa, Versicolor, and Virginica—with an accuracy of 100% on the test data.